

ENTERPRISE NETWORK: SECURITY ENHANCEMENT AND POLICY MANAGEMENT USING NEXT-GENERATION FIREWALL (NGFW)

Md. Taslim Arefin¹, Md. Raihan Uddin², Nawshad Ahmad Evan³, and Md
Raiyan Alam⁴

Daffodil International University, Dhaka, Bangladesh^{1,2,3}

Texas A&M University-Kingsville, USA⁴

{arefin}@diu.edu.bd¹ {mkraihan13}@gmail.com² {nawshadevan}@gmail.com³
{md_raiyan.alam}@students.tamuk.edu⁴

Abstract. Networks security is considered as a major task in network architecture. A network administrator had to focus and it is defined and demonstrated as the rules, plans, and procedures followed by a network administrator to protect the network devices from different threats and simultaneously like passive and active attacks form various vulnerable sources. Unauthorized users must be prevented from accessing the network. There are different types of threats that need to identify, explore and take a step for preventing it, the attacks are like DoS and DDos attacks, Aurora attacks, Malware Attack, Port Scanning, password sniffer, IP Spoofing, Session Hijacking, and Man-in-the-Middle Attacks, Cyberattacks. This could be done with the help of firewalls that secure the network from malicious attacks. This paper focused on strong policy and make incredible directions for averting mentioned attacks. Firewalls are one of the strongest hardware attachments to secure the zone of networking sectors like local large, Multinational, or Enterprise networks. The deployment of firewalls that enforce an organization's security policy are network devices. For this kind of tiresomeness, the concern of this paper is to enhance and develop network security like IPsec VPN, Strong Masquerades, Port forwarding, create a trusted zone on WAN and LAN side, etc based on the firewall by the execution of various tasks and different policies.

Keywords: Network Security · NGFW · Firewall · Enterprise Network.

1 Introduction

Now, in this modern world computer network system increases rapidly day by day. Therefore a strong security system is very much important to protect the network. In a computer network, users interface are threatened from all kind of attacks from hackers. So, the development of network systems is the most required step where unwanted attacks are rising significantly. A large number of

these dangers have turned out to be cunningly practiced assaults causing harm or submitting robbery. Moreover, government and business are massively dependent on the use of the internet, there are numerous advantages for using of internet. So, ensuring secure applications for this sector to prevent the risk. Firewall concepts and technologies are nowadays a major topic that should be a part of any information security curriculum. Firewalls are security devices or software tools that are used to apply by implementing security rules to filter inward and outward network traffic of an organization's security policy. However, commercial firewalls are nowadays designed to be used by professionals and are not usually very appropriate for the academic environment. That is the most commercial firewalls do not allow educators to suggest advanced hands-on lab exercises on firewall concepts and technologies and offer usually complex user interfaces to configure the firewalls and manipulate the filtering rules. For examples, [1] Low-level filtering rules is not to be allowed in commercial firewalls by manipulating the six flags of TCP, [2] also do not allow the lack of the ability to deal with the efficiency and the consistency of the filtering rules, [3] do not allow managing and viewing the table of stateful sessions, do not offer means to enhance the firewall packet processing performance by reordering the orders of the filtering rules and their rule-fields or using early packet rejection and acceptance mechanisms [4] [5] do not allow implementing rules to filter the application layers' contents (known as deep packet inspection (DPI)) and are unable to deal with common denial of service (DoS) attacks targeting the firewalls [6]. Consequently, commercial firewalls do not allow both the instructors to offer advanced hands-on lab exercises on firewalls, and the students to better anatomize advanced firewall concepts and technologies. An enterprise network is a network that helps to connect associate's PCs and related gadgets crosswise over offices and work group systems. An enterprise network decreases correspondence conventions, encouraging framework, and gadget interoperability, and also enhanced the interior and outside enterprise data management which is called a corporate network. An enterprise network is to eliminate different clients and work groups as the key motivation. All methods should be able to communicate with each other and provide and recover data. Also, physical processes and devices should be able to maintain and give favorable reliability, performance, and security. Enterprise computing models are developed of established enterprise communication protocols and strategies for this purpose, facilitating exploration and improvement. [7] on the other hand, an enterprise network includes the LAN/WAN based on departmental and operational supplies. [7] Since the improper security model of an enterprise network exists, this paper gives an overview of the different kinds of security attacks of an enterprise network and how to prevent these attacks. And finally gives a properly deployed security model. The objectives of this work are to publish and define the idea of attack and threat to a computer network, to highlight different allaying techniques used to complicate threats and attacks, to illustrate the procedure to implement the best security model for an enterprise network. [8] The various type of attack and different kinds of mitigation techniques are discussed in this paper. Finally implemented a secure network which

prevents those attack. Finally, this paper will introduce the importance of the network security of an enterprise network. Improper configuration of security in an enterprise network. Also, explain the newly deployed security model of an enterprise network. This newly deployed security model minimizes threats to give a better secure network. Also, this security-enhanced model can be resolved the future attack.

2 Related Work

A network admin can control the authorization of access in a network by deploying network security. To prevent the attack and ensure a safe and stable network, it is mandatory to exploit security. It can be easily maintained what services are allowed and what is not by using security policies in a firewall. Using a firewall for network security can be a sophisticated way to secure the service ports from the attackers, maintain valid user access for valid services, inspect the traffic and detect the harmful contents like malware, Trojans, and other computer worms. [7] [8] Nowadays security specialists are using Wireshark to capture packets and taking logs for analyzing the network system. Different kinds of encryption methods and security algorithms are used for the host to host communication. The communication technology is making the world smaller by enlarging and interconnecting the IP network. People from every stage are now adopting network security for creating a safe network environment of their data. The network security is now getting more priority for the scalable service and the research goes to analyze it. [8] To ensure a better security environment it is important to have security in all layers of the network. There are seven layers in the open system interconnection (OSI) model and we need to ensure security in all layers. In general, people think about only workstation security by applying security features in the application layer. To protect the data, it is important to exploit the security feature in all layers. [9] There can be any kind of malicious attack happen from any source when any workstation gets connected with the internet. There can be two kinds of attacks in general and those are ‘active attack’ and ‘passive attack’. In the active attack, the attacker will ambush the target directly and will try to modify the system or information. In another word, the passive attack will make the attacker breach the transit security of the network to analyze user data and afterward monitor. To avoid these, a network security engineer must apply some strong security framework. [10] As a next generation firewall ‘Fortinet’ has a tremendous contribution by delivering high performance, sophisticated security solutions, and integrated features. This is more capable to analyze the vulnerable traffic and a better option for the security engineers. [11] Security architecture added the feature like intrusion prevention system (IPS) which will analyze the outgoing and incoming traffic to the network. The firewall device needs to have this additional module to implement it. This feature analyzes the packet and filters the infected portion. [12] There are two kinds of firewall and these are hardware based and software based. Software based firewalls are generally installed in a host workstation and have so many drawbacks.

Hardware based firewalls are dedicated a device to operate security systems and it can be set up in any position of the network. This ensures better protection and solution than software-based firewalls. At present we have new kinds of firewalls which are a cloud-based firewall. These kinds of firewalls are easily accessed from remote and operated by specialized security professionals. [13]

3 Methodology

Most of the organizations found out the existing security system controls are reducing their effectiveness and preventing them from getting something done. An organization needs to share some information with the general user, customer, or newfangled business partners. If there are any kinds of security issue occur, what will be the business impact? For the outcomes of this kind of security issue, it needs to be developed an innovative, effective security program. So, the risk depends on the security system and security policy that the administrator applied to the network. Sometimes good security of an organization creates a good business opportunity because the customer, business partners want to be sure that their data is secure as new risks are introduced daily so the administrator needs to be aware of everything and should apply a proper firewall policy to prevent those attacks risks. The company needs to balance the security issue and business information while sharing company information can share information easily.[13]

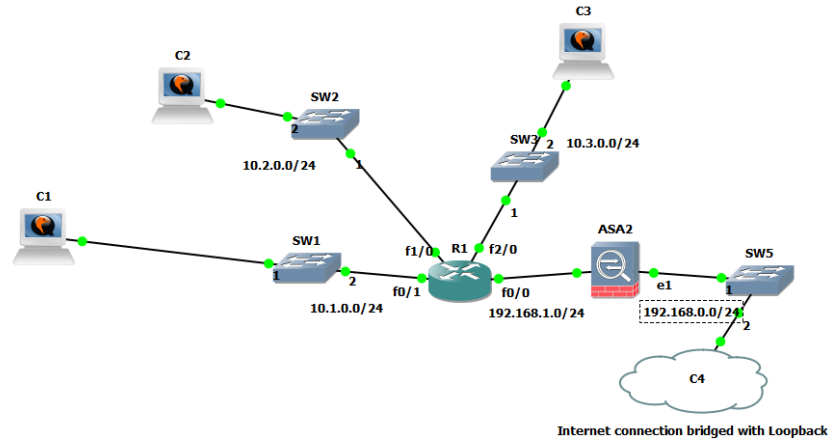


Fig. 1. Existing Network Model of an Enterprise Network.

3.1 IP Spoofing

Ip spoofing is to obtain unauthorized entree to the computers which are a kind of technique that attackers use. In this process, The hackers send illegitimate messages to the host computer IP as a trusted host. On another hand to generate traffic in the target computer and make it coming from a trusted host attacker engages with the unsuspected hosts, then the attacker floods the network. The way of transmitting packets with forged source addresses is known as IP spoofing. Which is straight linked to different kinds of networks malfunctioning like DDoS [14], [15]

3.2 Insider Intrusion

Insider intrusion is kind of unusual type of attack it is not like an external threat, A person who is authorized system access could be an insider intrusion attacker, So there may be less security against insider intrusion on most of the organization because most of the company focus on their external attacks, Insider attack is also recognized as insider threat.[16]

3.3 Denial of Service (DDoS)

The DDoS attack is distributed for a server, website, or another network resource by Multiple computer systems attack. The target system forces it to delay or even crash and even shut down as a flood of an incoming message, connection request, or malformed packets to the target. [17]

3.4 No protection against masquerades

A masquerade attack happens by using the fake identity of a network or system to acquire unofficial access to personal or organizational data. The attacker tries to steal user passwords and other confidential information. To prevent masquerade, attack some security features must be enabled for authorization and proper authentication.

3.5 Firewall trusted and untrusted network (LAN WAN)

In a firewall, the LAN is by default accepted as a secure network, and WAN is insecure. For that reason, security architectures named the LAN as trust and the WAN as untrusted. The trust network is defined as inside network and the security level is higher where the untrusted network is defined as outside and the security level is lower. It is allowed to initiate traffic from a higher security level to lower. In another way around traffic initiation from lower to higher works opposite. [18] Security measures are appropriate or not depend on the threat profile that helps us to understand network management for a self-connected LAN network, Here no need to have network management protocol encryption or special authentication for those protocols, the network administrator does

not want his network management protocols to traverse without the special authentication of internet protocols, So for any system first step is to identify the threat and then apply threat prevention policy to ensure security protection.

The proposed model may overcome the above all existing problems of an enterprise network. Here is those probably improved part of the model given below:

- Strong Authentication.
- High Data Security.
- Multilevel Protection.
- Network Traffic Encryption.
- No Insider Intrusion.
- Strong Masquerades.
- IP sec used.
- Port forwarding.
- Internet traffic filtering.
- Different web access policies for users.

4 Implementation

Here, we focused on the main things that we want to implement and enhance the security model of our proposed system of an enterprise network. The purpose of this security of an enterprise network to protect it's valuable data, file, document, and many other things from attackers, hackers, and other dangerous platforms. To prevent those attacks and enhance our security, we need proper authentication and authorization. In section 5.1 we will see what kind of tools are needed for this Process to implement in the enterprise networks and proper evaluation. In Section 5.2 we will see that the process of policy to apply like NAT, ACL, WAN and LAN zone, trust, and Untrusted network determine and various things. In section 5.3 we will see that port forwarding in the LAN side to secure our potential websites, server, different devices that are accessible by the internet with a major concern of security. In Section 5.4 we will see that how IPsec VPN policy is applied for higher security purposes and better authentication from vulnerability and packet spoofing is resolved from active and passive attacks. And at the end of this sector, we will see that Bandwidth utilization of this proposed system that we want to implement an enterprise network.

4.1 Devices and Appliances

To enhance our proposed model, we draw a network architecture of our desire system and make a topology in Graphical Network Simulator-3 (In Short GNS3). In this part, we took three cisco router and two Fortinet firewalls and some non-manageable switch connected with the Internet cloud separately. In the existing system, the ASA 8.42 firewall is used which is costly and expensive and also have some limitation. There are different types and better-secured firewalls in

our Internetworking world to secure the world of the internet now in this era. Some of these firewalls like Cisco ASA, Fortinet, Fortinet-Hyper version FortiGate firewall, Sophos firewall, Software firewall (Like ACL), Barracuda spam firewall, etc. are available. We will implement the FortiGate firewall in this system to enhance our security. This firewall has excellent features with flexibility and comfortability both graphically and CLI mode. Fortinet with the advanced version of FortiGate has various update versions that are user-friendly and highly secured to protect our LAN and WAN portion on the Internet. free Meter tools used for Bandwidth monitoring and bandwidth utilization of our existing system as well as how much improved in our proposed system to obtain better results and help to find out the effective results and also inefficiency measuring. This tool installed in host pc that means any change in bandwidth gives us the result of bandwidth utilization. This free meter tools are designed for the Windows operating system which is almost reliable for Bandwidth Monitoring and also for the process of evaluation. WinMTR is one of the free tools to lookup traceroute of a networking device connected with the Internet. Having a connection with the proper establishment with actual route from source to destination can easily give us traceroute hop to hop and also ping statistics of a network. This is also called Network diagnostics tools that are truly named after Matt's traceroute. A Network connection with the proper route in topology or a diagram facing any network difficulties or slow speed can be determined by the WinMTR. Fortinet-FortiGate hyper version is a Next-Generation firewall with advanced features and technology that is used for higher security in the LAN and WAN side as well as secure our network connection in the world of the internet. In my opinion for an enterprise network, the Fortinet firewall is the best solution to fulfill or requirement and meet up the challenge of privacy and protection. The firewall leads the maximum efficiency in both platforms (Virtualized and Hardware).

The devices and tools are used in this topology are given below:

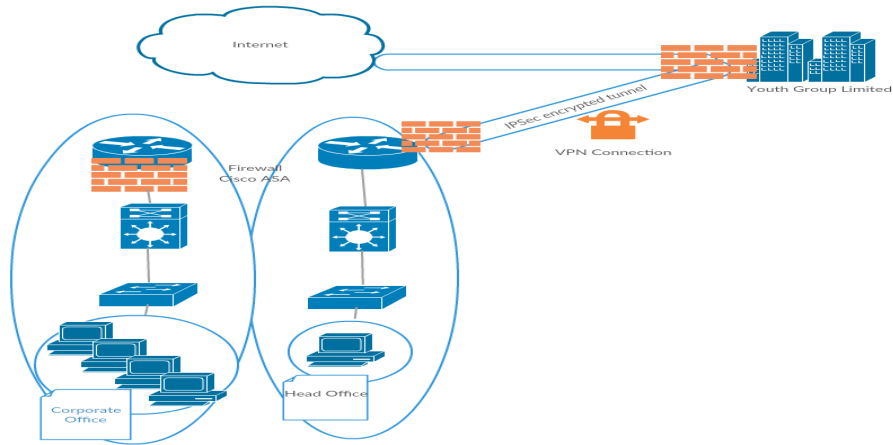
Network Design: The network or topology of our proposed system of an enterprise network is given below compared to an existing System we already showed it previously. In this topology, it's a scenario of an enterprise network where Corporate office and branch office connected to head office in the same network. In this topology, the internet cloud-connected with a Fortinet firewall for concern of security, privacy, and authentication. Then this Head office Firewall connected with another Fortinet firewall with encryption and using Internet Protocol Security (IPSec) VPN for higher security purposes along with the synchronization with routers, switches, L3 switches consecutively with end device (host PC). This is a common scenario of our proposed network to enhance the security of an enterprise network.

4.2 Implementation of firewall With the Integration of Different Policy

This section describes the different policies applied in Fortinet firewalls and various steps taken to remove the different attacks as those attacks mentioned before and how those attacks mitigate in FortiGate firewall. As in this Network

Table 1. List of Tools and devices for Implementation and application

Device Name	Specification	Installed Tools
Host PC – PC1 PC2 PC3	CPU: Core i4, 1.5 GHz	SecureCRT 8.5.2, WinMTR, FreeMeter
Cisco Switch	Layer 2 device, Non-Manageable	GNS3
Router c7200p	Layer3 device, Fast Ethernet port, Gigabit port, Maximum Speed accuracy	Cisco c7200 iOS, Wireshark
ASA 8.42 firewalls	Identity Firewall, Identity NAT configurable proxy ARP and route lookup	Cisco ASA 8.42 ios, ASDM bin
Firewall	Fortinet -FortiGate Hyper Versions	FortiGate VM, FortiGate-3140B
Metasploit Framework	Penetrating tools	Windows 10 OS

**Fig. 2.** Proposed Model for an Enterprise Network.

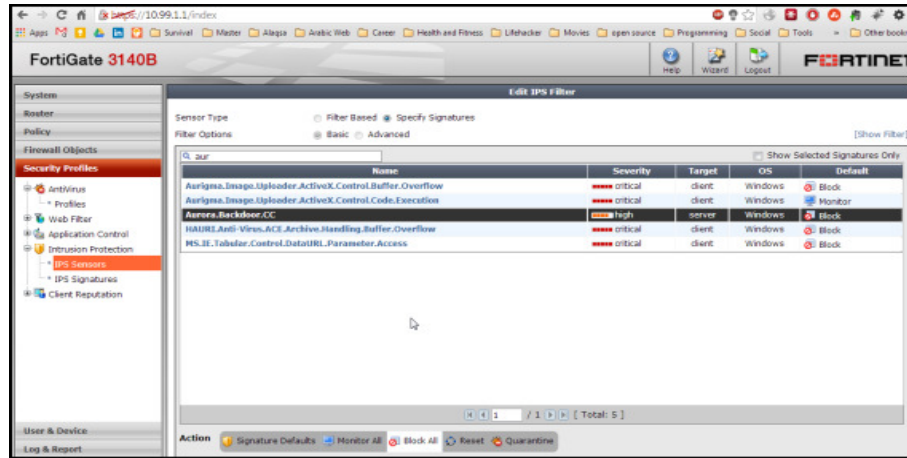


Fig. 3. Aurora Attack takes place IPS Page FortiGate-3140B.

topology where the whole network is scanning by the Wireshark software find the attacks before the action is needed from the firewall. For Implementing security both the results from the network with its scanning process are compared to show whether the reconnaissance. In Figure 5.2. Shows the FortiGate-3140B Firewall Graphical Mode of IPS page with an observation of the Aurora attack is detected and needs to prevent it. Aurora attack is highly harmful to the entire system. It targets s on the server host that takes place in the windows operating system as well as various random places that are essential for protecting data.

For Removing Attack and secure our data we need to protect HTTP, HTTPS, Telnet, SSH, SNMP port in the firewall as well as in the router. so It's essential to configure and monitor the firewall carefully. Setting Up the FortiGate Firewall and configured it properly. Figure (2) and Figure (3) shows that the IOs file configures in the VMware and established a connection Host PC and Virtual PC connected and setting IP and port In the for FortiGate Firewall in CLI Mode. When all the processes of Configure IP and port successfully configured then we can access the graphical Mode of FortiGate Firewall. The Assigned Ip in the port of Firewall having a Smooth Ping Statistics from Source to Destination.

4.3 Implementing Port Forwarding and Policy against Direct Internet Traffic

Port forwarding on router or firewall or any kind Device that controls or monitor traffic on the internet which allows a port address enter to it. Port forwarding plays an important role to secure vulnerable port of potential devices form attackers and hackers to keep safe our important data. With port forwarding, the internal port of a router or firewall along with an IP address can be changed where port number forwarded to unknown port for Incoming. This is a tricky way to secure and put privacy on the firewall to protect our Data. This firewall,

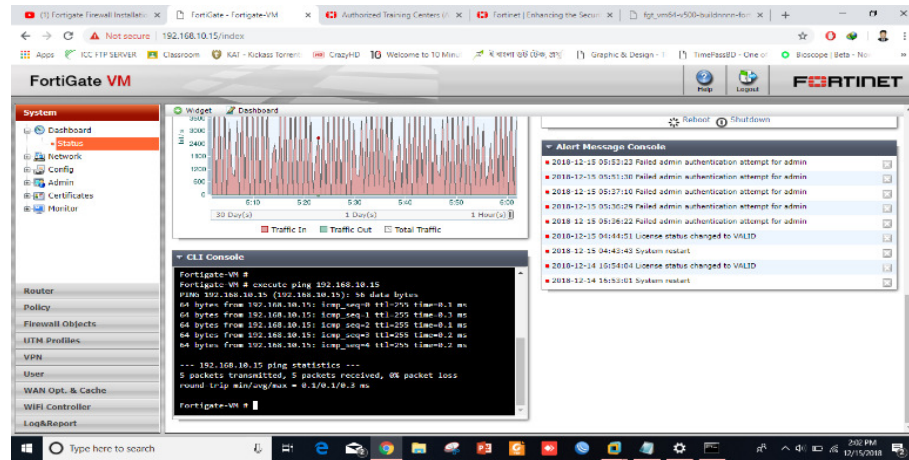


Fig. 4. Ping Statistics of FortiGate VM with Host PC.

we forwarded TCP port 21 to 8000 with the Private IP mapping with public IP as it gave form Internet Service provider. Port Forwarding or port mapping is part of Network Address Translation (NAT) where it is applicable. Direct Internet Traffic: Direct Internet traffic that a hacker sends malicious virus and different kinds of direct attack through third party software like uTorrent, Bit Torrent, and much other open-source software in our window operating system with the permission of access firewall without unknowing of mind.

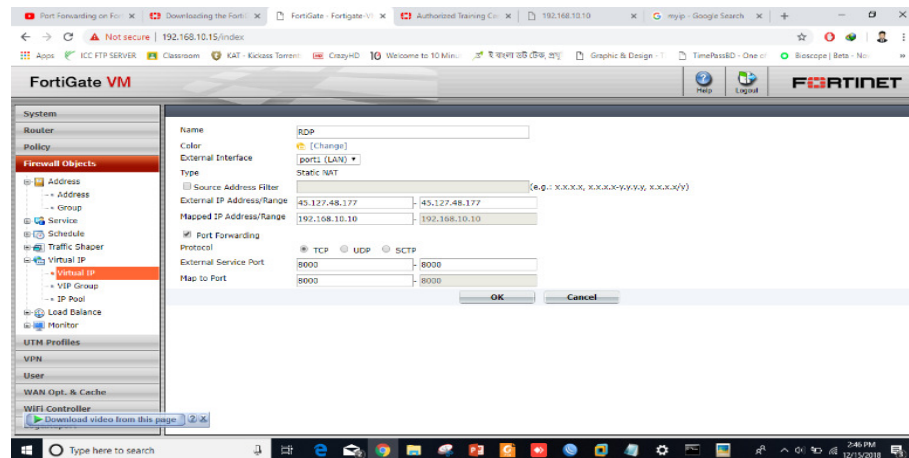


Fig. 5. Port Forwarding in FortiGate Firewall.

This an important problem and a security concern nowadays. It can be removed by our awareness when clicking any harmful link or installing any crack open-source software in our windows operating system. So, we create rules and policies like Objects (IP/Subnet, FQDN) create a FortiGate Firewall, and Introduced them in a particular declaration with these specific objects. These objects are blocked from the firewall were trusted and untrusted zone of LAN Sides. Some of the objects created FortiGate that's objects are restricted from firewalls when accessing some untrusted sites any suspicious websites. Then the policy applied for direct Internet Traffic to remove malicious attacks from different ports, open-source software, and create a strong zone for protecting valuable data from attackers. And the Traffic for direct internet monitoring shows in figure (4 / 5), in that monitoring gives us a concept or knowledge about direct traffic filtering in incoming and outgoing. We can understand from this figure how much problem occurs in direct internet traffic.

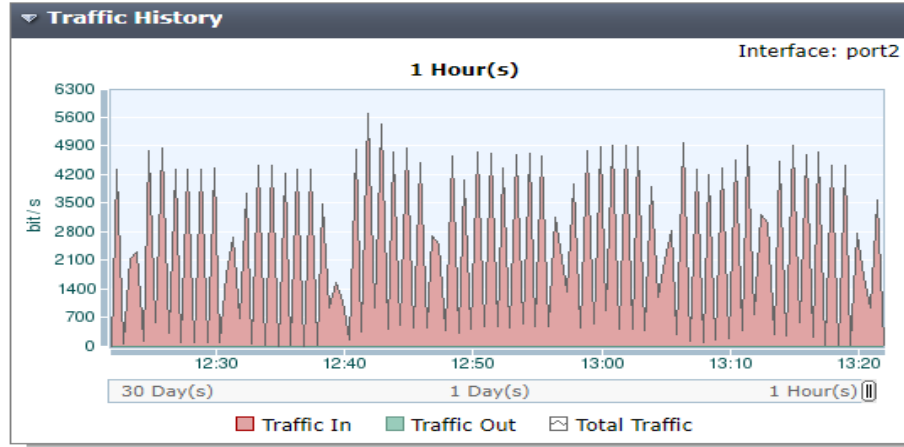


Fig. 6. Traffic History of Direct Internet traffic ingoing and outgoing.

4.4 Implementation of Internet protocol security (IPsec) VPN in Proposed Model:

The IPsec is used for Better security, IETF standard protocol. It's maintained the encryption process in sending data from one place to another. The IP layer is being secured when data is sending with an encryption process. A cryptographic key and also pre-shared keys use during the session. In our proposed model we applied IPsec VPN between two FortiGate firewalls which have a secure process in a big enterprise network to prevent Intrusion and different kinds of threats. For a session with end to end security, IPsec VPN provides the best TCP/IP layer security along with a higher level of authentication. In WAN port LAN port

where there is a route between source and destination, a pre-shared key is used for remote connection in FortiGate firewalls to another destination firewall with a virtual private connection is established. A tunnel is established Between head office to the corporate office with IPsec VPN for secure data internet passing and creates strong zone apart from the outside world as well as Internal network have several conditions of firewall policy when file-sharing or data.

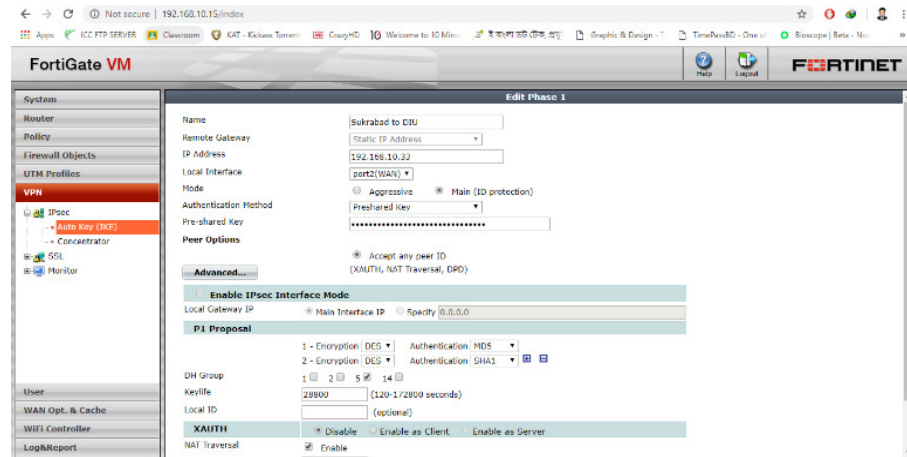


Fig. 7. VPN Configuration in local Interface.

Another IPsec VPN configures in the remote destination in the same way along with pre-Share keys encryption, a remote destination address, and Port (LAN) Selection. Figure (6 and 7) shows that the configuration of IPsec VPN for destination Host. The best path chooses to give encryption to make a session with local Host. And the firewall Policy applied in the FortiGate Firewall.

5 Results

In this section, we will evaluate the results of applying the different policies, rules, and various configurations in the Fortinet FortiGate Firewall. The following figures show the result of port forwarding, Protection against DoS Attack, Direct Internet traffic, Port Scanning, and IPsec VPN. This evaluation for the proposed System of an enterprise Network. Using the Metasploit tool for the penetrating attack in the FortiGate firewall. And this penetrating attack is done with various purposes like port scanning and Aurora attack for evaluation of log reports in the FortiGate firewall. Figure 8 shows that that penetrating attack by the Metasploit tool to detect Aurora attack, port scanning in the FortiGate firewall. In our findings in the exploit of a vulnerability in firewall using Internet explorer with local Host.

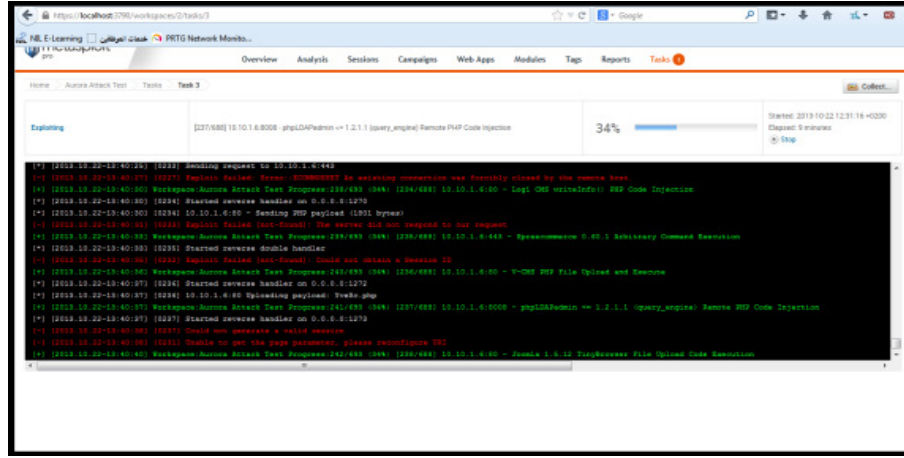


Fig. 8. Launching Aurora in Metasploit tools with port forwarding.

Some Object of Aurora attack with IP address and port in the FortiGate firewall. The vulnerability scan and log report Generate from FortiGate firewall is given below in figure 9

5.1 Performance Analysis and Evaluation:

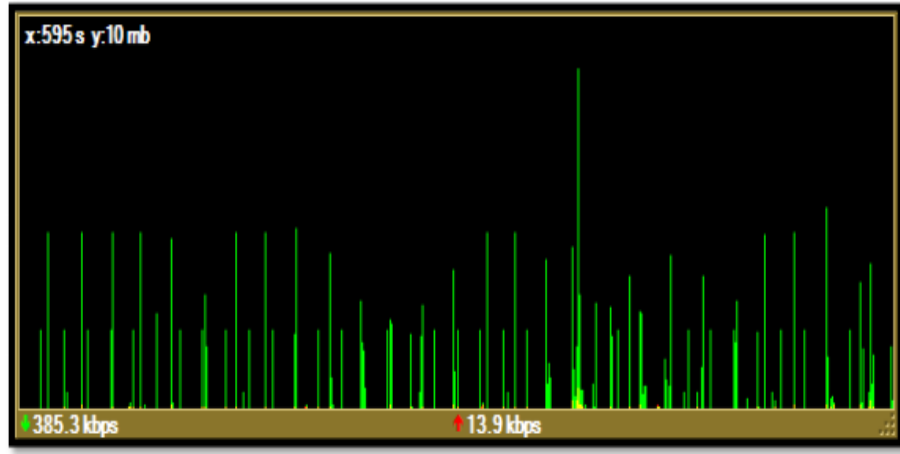


Fig. 9. Utilization of Bandwidth before applying the Proposed Model.

We use free meter tools to Monitoring and utilization of Bandwidth in the Proposed system of an enterprise Network compared to the existing system. One

more thing to say Fortinet firewall is comparatively cost-effective and Reliable than ASA (Adaptive Security Appliances). On the other Hand, ASA firewall is Costly though reliable when we think to design a network topology for an enterprise Network firstly we think how much cost-reducing when buying the network Equipment. An Effective Network topology not only helpful and Necessary for enterprise networks but also efficient and needful for a Company. When we applied different policies and rules in the firewall to protect our data from attackers or hackers then a matter a bandwidth also a concern with is. We use the free meter tools for measuring and mentoring bandwidth utilization when all the processes running in the firewall. Figure 11 shows bandwidth utilization before applying the proposed model and figure 12 shows the Bandwidth after using Firewall as the procedure already mentioned before in this paper.

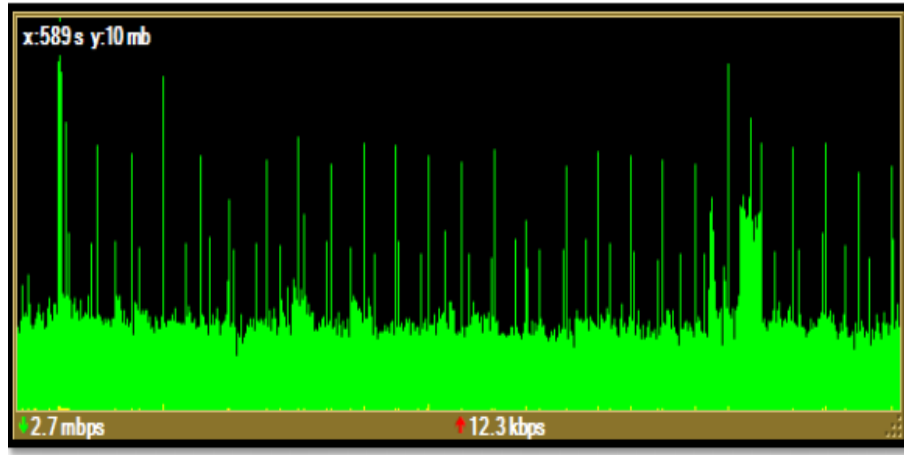


Fig. 10. Utilization of Bandwidth after using Firewall.

So comparatively Bandwidth utilization in our proposed monitoring System is better than the existing system considering maximum usages of maximum traffic. And we get CPU Usages is moderate than the existing System. If we differentiate attacks between the Proposed system and existing system in a chart of graphical view that it will be easier to Understanding and we will find out Anomaly of these system.

6 Conclusions

In this paper, we developed a network enhanced model for an enterprise network. Basically, an enterprise network includes a different device, network, equipment, and protocol. Different devices connected in a different layer. In layer two different devices like switches, printers are connected to the network. And layer three

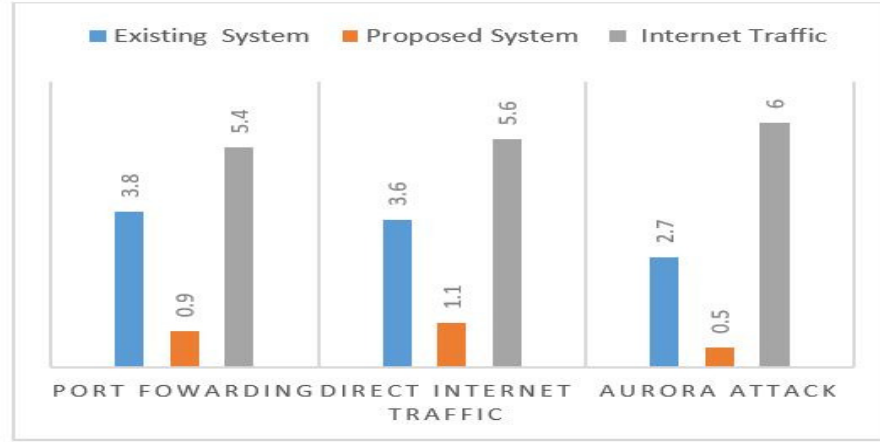


Fig. 11. Comparison between Existing and Proposed System

core switches like routers are connected to the network. These core switches are used to connect different branches in different networks. An enterprise network attacked by many threats like external and internal threats. Also, attacks came from different layers. To prevent this attack enterprise network uses different devices like VPN, Firewall. Security threats become a big challenge to create an enterprise network. The proposed network model in our paper also an enterprise network. This enterprise network is affected by different threats like IP spoofing, Phishing attack, DoS attack, spyware attack, etc. The proposed model gives a better solution to prevent the attack. As an enterprise network security issue is critical, we deployed firewalls and VPN to protect the network. In our paper, we keep studying our model to prevent different kinds of attacks. The results of our proposed security model prove that it has the ability to detect and protect the network from different kinds of attacks. But there is no guarantee that the proposed security model can detect the new attack. The proposed security model only detects some attacks to secure our enterprise network.

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